

Lecture V - Errors and Debugging

Programming with Python

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Checkpoint 2 — before the doors open

The first **40 minutes** are the checkpoint. It starts **now**.

- **Individual work:** no AI, no neighbors, no chat
- The **link and QR** are handed out in class. Open it and start
- ~6 short tasks: write code, trace code, fix a bug, answer a multiple choice

...

When you're done: menu → *Download* → *Download Python code* → upload the `.py` to the “**Checkpoint 2**” assignment on Moodle.

...

Note

The green  live checks are **provisional**. The final grading runs on our side. And take a breath: everything in it was rehearsed in the labs.

Episode 5: The 3-AM Checkout

After the checkpoint

Pens down. The checkpoint is behind you. Now the reason we're all here.

...

At 3 AM, running on his fourth energy drink, Kevin rewrote the **entire checkout** “to make it faster.” He went to bed very pleased. This morning two things are true: the till is a minefield of crashes, and the **health inspector** just called to announce a surprise visit.

...

So today the code learns to survive things going wrong. We read a **traceback**, catch failures with **try/except**, and let our own code **refuse** bad input.

Reading the Explosion

Kevin's 3-AM checkout, live

A customer typed `generous` into the tip box. Kevin's new checkout did this:

```
tip_text = "generous"
tip = float(tip_text)      # ✨
```

...

Python stopped and printed a **traceback**, a crash report. It looks scary, but it is the most useful thing on your screen:

```
Traceback (most recent call last):
  File "checkout.py", line 12, in <module>
    tip = float(tip_text)
ValueError: could not convert string to float: 'generous'
```

Read the traceback bottom-up

You read a traceback from the **bottom up**:

- the **last line** names *what* went wrong (`ValueError`) and gives a hint
- the lines **above** show *where*: the file, the line number, the guilty code

```
File "checkout.py", line 12, in <module>    ← WHERE it happened
    tip = float(tip_text)                  ← the exact line
ValueError: could not convert ...          ← WHAT went wrong (read
me first)
```

...

Two facts and you know where to go: **line 12**, and it's a `ValueError`. The scary wall of text is really just an address and a reason.

The big five (1)

A handful of exception types cover almost everything you'll hit. The first three:

- `ValueError` — right *type*, senseless *value*: `int("lots")`
- `TypeError` — the wrong type entirely: `"Bowl " + 9`
- `KeyError` — a dictionary key that isn't there: `table_map["C3"]`

...

Each one is Python refusing to guess. `int("lots")` has no sensible number; `"Bowl " + 9` mixes text and a number; `table_map["C3"]` asks for a key that was never added.

The big five (2)

The other two you'll meet constantly:

- `IndexError` — a list position past the end: `seats[9]` on a 3-item list
- `ZeroDivisionError` — dividing by zero: `88.00 / 0` when splitting a bill among no guests

...

You don't have to memorize all of Python's exceptions. Recognize these five on sight, and read the last line for the rest.

try / except: catch the fall

First, the happy path: text that **looks** like a number converts cleanly; text that doesn't, explodes. A **try / except** block runs the risky code and catches the failure instead of stopping the program:

```
print(float("5.50"))          # numeric text converts cleanly → 5.5

try:
    tip = float("generous")    # non-numeric text raises ValueError...
except ValueError:
    tip = 0.0                  # ...caught here, so the checkout keeps going

print(tip)
```

```
5.5
0.0
```

...

The risky line goes in the **try**; the recovery goes in the **except**. No crash. The program lands in the **except** and carries on.

Catch the specific type

Name the exact exception you expect. **except ValueError:** catches **only** value errors and lets anything else through:

```
try:
    tip = float(tip_text)
except ValueError:          # only this kind
    tip = 0.0
```

...

Warning

A **bare except:** catches *everything*, including your own typos (a misspelled variable would vanish silently). Catch the **specific** type, so real bugs still surface.

Predict: which explosion?

Kevin builds a table label. `guests` is the number `3`. What does the **last line** do?

```
guests = 3
label = "Party of " + guests
print(label)
```

a) prints `Party of 3` b) raises `TypeError` c) raises `ValueError`

...

Predict first. Commit to an answer before the next slide.

Answer: text and numbers won't mix

b) raises `TypeError` — `+` can glue two strings or add two numbers, but it refuses to mix a string and an `int`. The fix is to convert first (`str(guests)`):

```
guests = 3
try:
    label = "Party of " + guests
except TypeError as e:
    print("TypeError:", e)
```

```
TypeError: can only concatenate str (not "int") to str
```

⚡ Your turn — 10 minutes

Open the exercise (scan the QR or type the link):

beyondsimulations.github.io/Introduction-to-Python/notebooks/ex_05_a/



First **predict** what happens, then run it.

Debugging and Defending

The debugging loop

When something is wrong, you don't guess randomly. You follow a loop:

1. **Read** the traceback: it hands you the *what* and the *where*
2. **Reproduce** the crash: make it happen on demand
3. **Isolate**: narrow down to the smallest failing piece
4. **Fix**: change one thing, then run again

...

The inspector debugs the same way: find the broken thing, make it repeat, corner it, fix it. Panic is not a step.

`print()` — the beginner's flashlight

The simplest debugger is a well-placed `print()`. When you can't see what a value *is*, shine a light on it:

```
def line_total(qty, price):
    print("DEBUG qty =", qty, "price =", price)    # the flashlight
    return round(qty * price, 2)

print(line_total(3, 5.50))
```

```
DEBUG qty = 3 price = 5.5
16.5
```

...

Real debuggers with breakpoints exist and are wonderful, but a `print()` in the right spot solves most beginner bugs in seconds. Remove it once you've seen enough.

`raise` — when your code should refuse

Sometimes *your own* code should reject bad input on the spot. The inspector's rule is non-negotiable: **every price** ≥ 0 . `raise` throws an error deliberately:

```
def charge(amount):
    if amount < 0:
        raise ValueError("price cannot be negative")
    return amount

try:
    charge(-4.50)
except ValueError as e:
    print("refused:", e)
```

your code refuses it...
...on your terms

```
refused: price cannot be negative
```

...

A bad value stops *here*, at the door, instead of poisoning the books three screens later.

assert — a tripwire for invariants

An **assert** guards an *invariant*, a fact that must **always** hold. If it's true, nothing happens; if it's false, the program stops right there with an **AssertionError**:

```
subtotal = 9.60
assert subtotal >= 0, "subtotal went negative - bug upstream"
print("passed the tripwire:", subtotal)
```

```
passed the tripwire: 9.6
```

...

Think of it as a note to yourself, checked automatically: *“if this is ever false, something broke earlier. Stop before it spreads.”*

Try, then fall back

A friendly, common pattern: **try** the risky thing; if it fails, **fall back** to a sensible default so the program keeps moving.

```
def seats(text):
    try:
        return int(text)
    except ValueError:
        return 2                # a table seats two, unless told otherwise

print(seats("6"))              # a clean number → 6
print(seats("full"))           # nonsense → the safe default, 2
```

```
6
2
```

...

The caller never has to worry about a crash. Garbage in, sane default out.

Predict: what happens after the **except?**

`to_price` catches a bad value and returns `0.0`. After that, does the program reach the next line?

```
def to_price(text):
    try:
```

```
        return float(text)
    except ValueError:
        return 0.0

print(to_price("bad"))
print("checkout still running")
```

a) it never runs: the program already stopped b) it runs normally: the program carried on c) the `try` block runs a second time

...

Predict first. Commit to an answer before the next slide.

Answer: it carries on

b) it runs normally — that's the whole point of `try` / `except`. It *handles* the failure; once the `except` has run, control simply drops to the code after it, as if nothing had gone wrong:

```
def to_price(text):
    try:
        return float(text)
    except ValueError:
        return 0.0

print(to_price("bad"))           # 0.0 - recovered
print("checkout still running") # ...and we get here
```

```
0.0
checkout still running
```

⚡ Your turn — 10 minutes

Open the exercise (scan the QR or type the link):

beyondsimulations.github.io/Introduction-to-Python/notebooks/ex_05_b/



First **predict** what happens, then run it.

To the Lab

Tonight's episode

- Head to the lab notebook: [Episode 5 — The 3-AM Checkout](#)
- You'll read a traceback, write a price box that won't crash, fix Kevin's 3-AM receipt, make your code **refuse** negative prices with `raise`, and harden the whole checkout against a batch of poisoned orders
- It runs entirely in your browser: no setup, just click and code

...

! Important

Download your `.py` before you leave. Closing the tab without downloading loses your work, and downloading is exactly how you just handed in the checkpoint. Expect **red cells** in this lab: you'll cause errors on purpose. A red cell pauses everything below it. Fix it, and everything springs back.

Wrap-up

Three things to remember

1. A **traceback** is a crash report you read **bottom-up**: the last line names *what* broke, the lines above show *where*. That's an address and a reason, not a wall of noise
2. `try / except` catches a failure so the program **recovers** instead of stopping. Catch the **specific** type (`except ValueError`), never a bare `except` that also hides your own typos

3. Defend your own code: `raise` to refuse bad input (prices ≥ 0), `assert` to guard an invariant that must always hold, and **fall back** to a safe default when a conversion fails

...

Note

Next time — Episode 6 starts with **Checkpoint 3**, which sweeps everything from Episodes 1–5, so keep this notebook and the last four close. Then someone new walks into the shop, uncaps a marker, and writes one question on the whiteboard. **Part II begins.**

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

Note

Nothing new here, but these are still great books to start with!

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For more, see the [literature list](#) of this course.